

E-Tech Social Hub

Project Overview & Technology Report

What is this website?

E-Tech Social Hub is a full-stack web application built for an Electronics/Tech (E-Tech) department's social media club. It serves as a central hub for the department's various student clubs and communities — showcasing clubs, members, announcements, achievements, event activities, a photo gallery, and a live feed of social media posts. The site also includes user registration/login, a personal profile system, an achievement-submission workflow, and lightweight gamification (XP points, referral codes, and hidden "easter eggs") to encourage student engagement.

Core Features

- **Clubs directory** — browse all department clubs with dedicated profile pages (via dynamic *[slug]* routes).
- **Members** — club member listings with roles, bios, and social links (LinkedIn, GitHub, Instagram).
- **Announcements** — pinned/tagged department and club news feed.
- **Achievements** — a showcase of student/club accomplishments, plus a public submission form for students to nominate their own achievements for review.
- **Gallery** — event photo gallery with a lightbox viewer.
- **Activities** — listing of technical/social events by type, category, and date.
- **Social media feed** — embedded posts pulled from club social channels.
- **Authentication & profile** — JWT-based sign up/login, editable user profile, referral codes.
- **Gamification** — XP points and hidden "easter egg" discoveries surfaced through a widget and a secret page.
- **Live stats** — animated counters for total clubs, members, events, and achievements pulled from the API.

Frontend Technology

The frontend is a modern React application built with Next.js (App Router) and TypeScript, styled with Tailwind CSS and animated with Framer Motion.

Aspect	Technology
Framework	Next.js 16 (App Router, React Server/Client Components)
UI Library	React 19
Language	TypeScript
Styling	Tailwind CSS v4 (with PostCSS)
Animation	Framer Motion (page transitions, scroll reveals, floating elements)

Data Fetching	TanStack React Query (server-state caching) + Axios (HTTP client)
Icons	lucide-react
Utilities	clsx (conditional class names)
Build/Compiler	React Compiler (babel-plugin-react-compiler), ESLint 9

Frontend Structure

- **src/app/** — route pages: home, about, clubs (+ dynamic club slug pages), members-facing achievements, announcements, activities, gallery, social-media-club, auth, profile, submit-achievement, and a secret/easter-egg page.
- **src/components/** — reusable UI: Navbar, Footer, ClubCard, MemberCard, SocialPostCard, AnimatedCounter, ParticleBackground, ScrollReveal, Lightbox, GamificationWidget, EasterEggManager, InteractiveCursor, ThemeProvider, LoadingSpinner.
- **src/lib/** — api.ts (Axios wrapper functions for backend endpoints) and auth.tsx (auth context/provider), providers.tsx (React Query provider setup).

Backend Technology

The backend is a Node.js REST API built with Express, using SQLite as an embedded, file-based database — a lightweight choice well suited to a single-department club site.

Aspect	Technology
Runtime	Node.js
Web Framework	Express 4
Database	SQLite via better-sqlite3 (WAL journal mode, foreign keys enabled)
Authentication	JSON Web Tokens (jsonwebtoken) + bcryptjs password hashing
Security Middleware	Helmet (HTTP headers), CORS, express-rate-limit (200 req/15 min)
File Uploads	Multer (served statically from /uploads)
Config	dotenv (.env)
IDs	uuid
Dev Tooling	nodemon

Backend Structure

- **src/server.js** — Express app setup: security middleware, rate limiting, static file serving, route mounting, error handling.
- **src/models/database.js** — SQLite schema definition/initialization (clubs, members, announcements, achievements, gallery, activities, social_posts, achievement_submissions, department_info, users, referrals).
- **src/routes/** — REST endpoints: auth, users, clubs, members, announcements, achievements, gallery, activities, socialPosts, submissions, department, stats.
- **src/middleware/auth.js** — JWT verification middleware for protected routes.
- **src/seeds/seed.js** — database seeding script for initial demo data.

Database Schema (SQLite)

- **clubs** — name, slug, tagline, description, branding (color/icon), social links, member count.
- **members** — per-club member profiles linked via club_id.
- **announcements** — title, content, tag, pinned/published flags, optional club association.
- **achievements** — title, description, category, proof/image URLs, featured flag.
- **achievement_submissions** — user-submitted achievements pending review (status: pending/approved/rejected).
- **gallery** — event photos with category/year metadata.
- **activities** — events with type/category/date/location.
- **social_posts** — embedded social media posts per platform.

- **department_info** — department-wide description, vision, mission, highlights, stats.
- **users / referrals** — accounts (with XP points, easter eggs found, referral codes) for auth and gamification.

Architecture Summary

The project follows a classic decoupled client-server architecture: a Next.js frontend (deployed separately, e.g. on Vercel) communicates with an independent Express + SQLite REST API over HTTP (Axios/React Query), with CORS restricting access to the configured frontend origin and JWT securing authenticated endpoints.

Layer	Technology Stack
Frontend	Next.js 16 + React 19 + TypeScript + Tailwind CSS + Framer Motion
State/Data	TanStack React Query + Axios
Backend API	Node.js + Express 4 + JWT Auth
Database	SQLite (better-sqlite3, WAL mode)
Security	Helmet, CORS, rate limiting, bcrypt password hashing

Generated automatically from the project source code.